

Lecture 31 — Rank, column space, null space, and basis connections

Wed Nov 4

Last updated: August 18, 2026.

The class outline for this day will be posted before class.

Reading: Boyd plus selected portions of [Strang §3.5](#). Related objectives: [Lectures](#).

Learning objectives

By the end of class, students should be able to:

1. Interpret column space and null space through $Ax = b$ and $Ax = 0$.
2. Determine rank from pivots or independent columns.
3. Find simple bases for column or null spaces.
4. Connect rank to consistency, independence, and dimension.
5. Use subspace, basis, dimension, rank, column space, and null space correctly.